EXCEL FORMULAS & FUNCTIONS

Complete learning handbook | Lessons 42-46 | Fundamentals, SUM, AVERAGE, COUNT, COUNTA, MAX & MIN

Purpose: A practical reference covering all concepts, sample datasets, formulas, exact Excel steps, outputs, edge cases, and related functions introduced in our lessons. Practice assignments are intentionally excluded.

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Lesson	Coverage
42	Formula fundamentals, formula vs function, operators, environment, errors
43	Function categories, insertion methods, SUM and edge cases
44	AVERAGE, row/column averages, blanks, zeros, text, AVERAGEA
45	COUNT, COUNTA, COUNTBLANK, filtered and distinct counts
46	MAX, MIN, dates, negatives, finding associated labels, MAXIFS/MINIFS

Workbook: Excel_Complete_Practice.xlsx. Add one worksheet per lesson using the names shown. Excel UI paths reflect common Windows desktop versions; labels can vary slightly.

LESSON 42 | Formulas & Functions Fundamentals

Why use formulas?

Formulas calculate results, update when referenced inputs change, reduce manual arithmetic, and make calculations reproducible. For quantity 5 in B2 and price 200 in C2, enter =B2*C2 in D2; output is 1000.

Formula vs function

Formula	Function
User-built expression such as =A1+A2+A3	Excel predefined calculation such as =SUM(A1:A3)
Can contain operators, references, and functions	Can be included within larger formulas
=SUM(A1:A5)*10% is a formula that uses a function	SUM is the function within that formula

Formula environment

Item	Meaning / example
Cell / reference	A1 identifies a cell; B2 is another cell
Formula bar	Displays or edits the selected cell contents
Name Box	Shows the selected reference; can navigate to a cell or range
Range	A1:A10 means cells A1 through A10
= sign	Starts a standard Excel formula
Arguments	Inputs supplied to functions, e.g. SUM(A1:A5)

Common operators and precedence

Operator	Function	Example
+ / -	Addition / subtraction	=A1+B1 / =A1-B1
* / /	Multiplication / division	=A1*B1 / =A1/B1
^	Power	=A1^2
%	Percent	=A1*10%
&	Join text	=A1&" "&B1
=, >, <	Compare values	=A1>B1 returns TRUE or FALSE

Order of operations: =10+5*2 returns 20, while =(10+5)*2 returns 30. Use parentheses when calculation order matters.

First formula: worksheet 42_FormulaFundamentals

Cell	Entry
A1	100
A2	200
A3	300
A4	=A1+A2+A3 (result: 600)

1. Create the worksheet and enter the values in A1:A3.

2. Select A4, type =A1+A2+A3, and press Enter.

3. Change A1 from 100 to 500: A4 becomes 1000 if calculation is Automatic. Restore A1 to 100.

4. To edit a formula, select its cell and press F2 (or use the formula bar); press Enter to save or Esc to cancel.

Hardcoded vs referenced: =100+200 remains fixed, while =A1+A2 responds when inputs change. Prefer references for changing values.

Common formula errors

Error	Meaning
#DIV/0!	Division by zero or no numerical divisor
#VALUE!	Incompatible argument or value type
#REF!	Invalid cell reference
#NAME?	Unrecognized name or function
#NUM!	Invalid numerical calculation
#SPILL!	Dynamic-array result is blocked

Calculation setting: Formulas generally recalculate automatically. Check Formulas > Calculation Options > Automatic if results do not update.

LESSON 43 | Function Types, Insertion & SUM

Main function categories

Category	Examples
Mathematical	SUM, ROUND, ABS
Statistical	AVERAGE, COUNT, MAX, MIN
Logical	IF, AND, OR
Text	LEFT, RIGHT, MID, LEN
Date and Time	TODAY, NOW, YEAR
Lookup & Reference	XLOOKUP, VLOOKUP, INDEX, MATCH
Financial	PMT, FV, PV
Information	ISNUMBER, ISTEXT, ISBLANK

Three ways to insert a function

1. Direct entry: click an output cell, type =SUM(A1:A5), press Enter.

2. Insert Function: select output cell > click fx beside formula bar > search SUM > choose function > choose argument range > OK.

3. AutoSum: click directly below a numeric column > Home > AutoSum (Σ), or Alt+= > verify suggested range > Enter.

SUM syntax and sample data

=SUM(number1,[number2],...)

Required first argument can be a value, reference, or range. Further arguments are optional. The square brackets are documentation notation, not characters to type. Some Excel regional settings separate arguments with semicolons.

Worksheet: 43_SUMFunction. Enter the following data in A1:D6.

Product	January	February	March
Pizza	12000	15000	18000
Burger	8000	9500	11000
Sandwich	5000	6500	7000
Pasta	10000	12000	14000
Fries	4000	5000	6000

SUM variants and outputs

Formula	Result / explanation
=SUM(10,20,30)	60: direct arguments
=SUM(B2:B6)	39000: January column total
=SUM(B2,B4,B6)	21000: selected January cells
=SUM(B2:B6,D2:D6)	95000: January plus March
=SUM(B2:D2)	45000: Pizza across three months
=SUM(E2:E6)	143000: all product totals (when E contains row totals)

Row and column calculations

1. Type Total Sales in E1. In E2 enter =SUM(B2:D2).

2. Drag the fill handle from E2 down to E6. Relative row references update automatically.

3. In B7 type =SUM(B2:B6), in C7 type =SUM(C2:C6), in D7 type =SUM(D2:D6).

4. In E7 type =SUM(E2:E6); avoid including E7 itself in its sum.

Product	Row total
Pizza	45000
Burger	28500
Sandwich	18500
Pasta	36000
Fries	15000
January total	39000
February total	48000
March total	56000
Grand total	143000

SUM behaviour and limitations

Input in referenced range	SUM treatment
Numbers, including zero and negatives	Included
Blank cells	Ignored
Text and logical values stored in referenced cells	Ignored
Error values	Usually cause an error result
Text numeral passed directly as argument	May be coerced, e.g. =SUM("100",200) yields 300

SUM vs +: With A1=100, A2=Apple, A3=200: =SUM(A1:A3) returns 300; =A1+A2+A3 returns #VALUE!. With values 1000, 2000, -500, SUM returns 2500.

Filtered rows: SUM ordinarily includes filtered-out values. =SUBTOTAL(109,B2:B101) sums visible filtered records and excludes manually hidden rows. Avoid double-counting grand totals.

LESSON 44 | AVERAGE Function

What AVERAGE calculates

=AVERAGE(number1,[number2],...)

AVERAGE calculates arithmetic mean: sum of numerical observations divided by their numerical count. Example =AVERAGE(80,90,70) returns 80.

Use worksheet 44_AVERAGEFunction

Reuse the five-product monthly sales dataset from Lesson 43 in A1:D6. In B7 enter =AVERAGE(B2:B6), then copy across through D7.

Month	Formula	Expected average
January	=AVERAGE(B2:B6)	7800
February	=AVERAGE(C2:C6)	9600
March	=AVERAGE(D2:D6)	11200

Average across each product row

1. Type Average Sales in E1.

2. In E2 enter =AVERAGE(B2:D2).

3. Drag E2 down through E6; the referenced row changes automatically.

Product	Average sales
Pizza	15000
Burger	9500
Sandwich	6166.67
Pasta	12000
Fries	5000

Other forms

Formula	Result
=AVERAGE(10,20,30)	20
=AVERAGE(B2,B4,B6)	7000
=AVERAGE(B2:B6,D2:D6)	9500
=SUM(A1:A3)/COUNT(A1:A3)	Same as AVERAGE(A1:A3) for numerical data

Blank, zero, text and negative examples

Referenced cells	AVERAGE result
10,20,blank,30	20 (blank excluded)
10,20,0,30	15 (zero included)
10,20,Absent,30	20 (text excluded)
100,-20,40	40 (negatives included)
Only text or empty cells	#DIV/0! (no numeric observations)

Direct arguments: =AVERAGE("10",20) can coerce direct numeric text and return 15; text stored in referenced cells is generally ignored. AVERAGE includes filtered-out rows. AVERAGEA counts text in references as 0 and TRUE/FALSE as 1/0; it is not equivalent to AVERAGE.

LESSON 45 | COUNT, COUNTA & Counting Limitations

Definitions

=COUNT(value1,[value2],...)

=COUNTA(value1,[value2],...)

COUNT counts numerical values in referenced cells. COUNTA counts cells that are not truly empty, including text, errors, and formulas returning an empty string.

Worksheet 45_COUNT_COUNTA: enter in A1:E7

Employee ID	Name	Age	Department	Salary
E101	Arun	25	IT	40000
E102	Sneha	28	HR	45000
E103	Ravi	(blank)	IT	50000
E104	Priya	30	Finance	(blank)
E105	Kiran	0	(blank)	35000
E106	Meena	32	HR	42000

Enter cells marked (blank) as genuinely empty cells, not the literal word “blank”.

COUNT and COUNTA outputs

Range	COUNT	COUNTA
A2:A7: Employee IDs	0	6
B2:B7: Names	0	6
C2:C7: Ages	5	5
D2:D7: Department	0	5
E2:E7: Salary	5	5

What each function counts

Cell value	COUNT	COUNTA
Number or 0	Yes	Yes
Text or text ID	No	Yes
Valid numeric Excel date	Yes	Yes
TRUE/FALSE in referenced cell	No	Yes
Error value	No	Yes
Space character	No	Yes
Formula returning ""	No	Yes
Truly empty	No	No

Direct arguments: =COUNT(10,20,TRUE,"30") returns 4 because COUNT may coerce logical and numeric-text constants given directly. Referenced text/logical values are handled differently.

COUNTA limitations and related functions

COUNTA counts a cell containing a space or formula ="" even though it may appear blank. COUNTBLANK counts genuinely empty cells and cells whose formula returns ""; consequently COUNTA and COUNTBLANK can both count the same formula-blank cell.

Question	Formula	Expected result
How many employees?	=COUNTA(A2:A7)	6
How many numerical salaries?	=COUNT(E2:E7)	5
How many blank salaries?	=COUNTBLANK(E2:E7)	1
Average populated salary	=SUM(E2:E7)/COUNT(E2:E7)	42400
Count numerical age and salary cells	=COUNT(C2:C7,E2:E7)	10
Count IDs and names	=COUNTA(A2:A7,B2:B7)	12

Counting distinct or visible values

COUNTA counts duplicates too. If A1:A4 contains Pizza, Burger, Pizza, Sandwich, =COUNTA(A1:A4) returns 4, whereas =COUNTA(UNIQUE(A1:A4)) returns 3 in versions supporting UNIQUE.

COUNT and COUNTA ordinarily include filtered-out rows. Use =SUBTOTAL(102,C2:C100) for visible numeric count and =SUBTOTAL(103,A2:A100) for visible nonempty count; these function numbers also exclude manually hidden rows.

LESSON 46 | MAX & MIN Functions

Definitions and syntax

=MAX(number1,[number2],...)

=MIN(number1,[number2],...)

MAX returns the highest numerical value, MIN the lowest. They return values rather than labels of the associated records.

Worksheet 46_MAX_MIN

Reuse the product-month sales data in A1:D6 from Lesson 43.

Month	MAX formula	Highest	MIN formula	Lowest
January	=MAX(B2:B6)	12000	=MIN(B2:B6)	4000
February	=MAX(C2:C6)	15000	=MIN(C2:C6)	5000
March	=MAX(D2:D6)	18000	=MIN(D2:D6)	6000

Highest and lowest per product

1. Enter Highest Sales in E1 and Lowest Sales in F1.

2. In E2 enter =MAX(B2:D2); in F2 enter =MIN(B2:D2).

3. Copy E2:F2 down through row 6.

Product	Highest	Lowest
Pizza	18000	12000
Burger	11000	8000
Sandwich	7000	5000
Pasta	14000	10000
Fries	6000	4000

Additional MAX and MIN examples

Formula	Output
=MAX(10,50,30)	50
=MIN(10,50,30)	10
=MAX(B2,B4,B6)	12000
=MIN(B2,B4,B6)	4000
=MAX(B2:B6,D2:D6)	18000
=MIN(B2:B6,D2:D6)	4000
=MAX(B2:B6)-MIN(B2:B6)	8000

Data types and edge cases

Case	MAX	MIN
100, blank, 300, 200	300	100
100, Apple, 300, 200	300	100
100, 0, 300	300	0
-100, -50, -200	-50	-200
Only text / blanks	0	0

MAX/MIN ignore text, blanks and logical values in referenced cells, but include zero and negatives. Referenced errors normally propagate. If no numeric values exist, MAX and MIN return 0 (unlike AVERAGE, which returns #DIV/0!).

Dates: latest and earliest

Valid Excel dates are numeric serials. With 10-Jan-2023, 15-Mar-2024, 05-Jun-2022 in B2:B4: =MAX(B2:B4) gives 15-Mar-2024; =MIN(B2:B4) gives 05-Jun-2022. Format result cells as Date.

Finding the label associated with highest / lowest value

=INDEX(A2:A6,MATCH(MAX(B2:B6),B2:B6,0))

Returns Pizza for January sales. Substitute MIN(B2:B6) to return Fries. MATCH finds the position; INDEX returns the label. When maxima/minima tie, the first match is returned.

Filtered and conditional maximum/minimum

Requirement	Formula
Visible maximum	=SUBTOTAL(104,B2:B100)
Visible minimum	=SUBTOTAL(105,B2:B100)
Max for South region	=MAXIFS(B2:B100,A2:A100,"South")
Min for South region	=MINIFS(B2:B100,A2:A100,"South")

Ordinary MAX/MIN include filtered-out rows; the SUBTOTAL function numbers above exclude filtered and manually hidden rows. MAXIFS/MINIFS apply criteria and are introduced here for awareness.

Final function reference

Function	Main use	Key detail
SUM	Add numerical values	Ignores text in referenced ranges
AVERAGE	Arithmetic mean	Ignores blank; includes zero
COUNT	Count numeric cells	Text IDs are not counted
COUNTA	Count nonempty cells	Counts spaces, errors and =””
MAX	Highest value	Returns 0 if no numbers
MIN	Lowest value	Zero and negatives are valid

Review checklist: Verify the correct range, avoid including totals in source ranges, distinguish blanks from zero, distinguish text from numeric values, and check whether filtered rows should be included. This handbook contains no practice assignments.